

VENKATESH PENSALWAR

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PROFESSIONAL SUMMARY

Generative AI Engineer and Python Backend Developer with nearly 2 years of experience building production-grade Large Language Model (LLM) applications, Agentic AI systems, and machine learning automation solutions for enterprise environments. Proven expertise in Retrieval-Augmented Generation (RAG), LangChain, LangGraph, multi-agent architectures, FastAPI microservices, and cloud-native AI deployment on AWS. Skilled in OpenAI GPT-4, AWS Bedrock (Claude), prompt engineering, vector databases, LLM guardrails, and hallucination mitigation.

TECHNICAL SKILLS

Generative AI and LLMs: Python, Large Language Models (LLMs), OpenAI API (GPT-4), AWS Bedrock (Claude), LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, Autonomous Agents, Prompt Engineering, Function Calling, Tool Calling, FAISS, ChromaDB, Vector Databases, Embeddings, Semantic Search, Structured Output Validation, LLM Guardrails

Backend and APIs: FastAPI, Django, Flask, REST APIs, Microservices, PostgreSQL, MySQL, Pydantic, JWT Authentication, SQL, SQLAlchemy

Cloud and DevOps: AWS (EC2, S3, ECS Fargate, Bedrock, RDS, IAM, CloudWatch), Snowflake, Docker, Git, GitHub Actions, Linux

Tools: Postman, Swagger/OpenAPI, Streamlit, VS Code, React.js (Basics)

PROFESSIONAL EXPERIENCE

AI Engineer | Generative AI

Oct 2024 - Present

64 Squares LLC, Pune, India

- Architected multi-agent Agentic AI workflows using LangChain, LangGraph, and FastAPI to automate enterprise decision routing and clinical triage across 9+ modules and 90+ adaptive questions.
- Engineered core backend systems for Virtual Pain Clinic (VPC) using AWS Bedrock (Claude), enabling AI-driven symptom screening, adaptive question flows, automated ICD-10 code generation, and structured medical documentation.
- Built Hyundai AI Sales Assistant using LangGraph multi-agent workflows, persistent state management, and intelligent query reformulation over Snowflake SQL for real-time vehicle discovery and pricing comparison.
- Implemented custom Retrieval-Augmented Generation (RAG) pipelines combining FAISS-based vector search, text embeddings, and contextual retrieval to improve response accuracy and reduce model hallucinations.
- Developed robust LLM guardrails using JSON Schema validation, structured outputs, and business rule enforcement to ensure reliable enterprise-grade AI responses.

Python Full Stack Developer

Aug 2023 - May 2024

FirstBit Solution, Pune, India

- Developed scalable backend microservices using Django, Flask, and FastAPI with JWT authentication and role-based access control (RBAC); designed RESTful APIs with modular service-layer architecture.
- Optimized PostgreSQL and MySQL database performance through indexing strategies, query tuning, and caching; maintained College Management System covering student records, attendance, and dashboards.

KEY PROJECTS

Virtual Pain Clinic (VPC) | Agentic AI Healthcare Platform

Python, FastAPI, LangChain, LangGraph, AWS Bedrock, PostgreSQL (RDS), React, Three.js, Docker

- Engineered a production-grade Agentic AI intake system with autonomous Screening and Intake Agents managing decision-tree branching across 90+ adaptive clinical questions and 9+ clinical modules.
- Integrated interactive 3D pain mapping (React + Three.js) for anatomical metadata capture and structured ICD-10 code generation via AWS Bedrock (Claude) multi-agent reasoning workflows.

Hyundai AI Sales Assistant | Agentic Conversational AI

Python, FastAPI, LangChain, LangGraph, GPT-4, Snowflake, Streamlit

- Built an Agentic AI sales assistant using LangGraph multi-agent workflows, state management, autonomous tool-calling agents, and Snowflake SQL retrieval for real-time automotive variant comparison.
- Designed hybrid RAG pipelines merging structured SQL retrieval with LLM reasoning; built FastAPI streaming endpoints for low-latency, high-throughput conversational interactions.

EDUCATION

Master of Computer Applications (MCA)

Savitribai Phule Pune University

2021 - 2023

CGPA: 8.89 / 10

B.Voc in Software Development

Savitribai Phule Pune University

2018 - 2021

CGPA: 10.00 / 10

CERTIFICATIONS

Databricks Certified Generative AI Engineer Associate (2025) | Python Full Stack Development - FirstBit Solutions | Pro Python Certification - TechnoWebTech Pvt. Ltd.